

Trainings ▾

General Information

This section presents the various winter fuel issues and methods of dealing with them.

Two winter fuel handling issues, wax and ice, have annoyed diesel operators for years. There is no solution to either of these problems that is ideal for all situations, but the better the problem is understood, the less difficult the process of finding a solution becomes. Determining whether a low power complaint is due to a fuel filter plugging complaint is fairly simple: replace the fuel filter with a new filter. If this allows the vehicle to operate normally even for a short period of time, then obviously something in the fuel is plugging the filter and causing the complaint. A simple way of determining whether the filter plugging is caused by wax or ice is to bring the plugged filter into a warm shop, drain out the liquid fuel, place the filter upside down on a piece of paper or in a shallow pan, and allow the filter to warm to room temperature. If there is ice in the filter, it will melt and run out of the filter and the water on the paper or in the pan will be obvious. Most petroleum wax, on the other hand, will **not** melt at room temperature. To speed the analysis process, the filter can be cut open and spread out. Once the cause of the low power complaint is determined, a logical solution can be chosen.

Fuel Wax

All middle (or intermediate) distillate fuels, such as jet fuels, heating fuels, and diesel fuels, contain paraffin wax. Paraffin wax is a solid, crystalline mixture of straight-chain or normal hydrocarbons melting in the approximate range of 40 to 60°C [104 to 140°F]. This paraffin wax occurs naturally in the crude oil from which fuel oils are distilled. The wax content of a distillate fuel varies greatly, depending on the crude oil from which the fuel is produced and in the processing of the fuel. Generally, higher boiling distillate fuels, such as U.S. Number 2-D diesel fuel, have a higher concentration of paraffin wax than lower boiling distillate fuels, such as jet fuel.

Because of the strong relationship between temperature and solubility of wax, wax separation is a problem in handling and using diesel fuel during cold weather. As fuel cools, a temperature is reached at which the soluble paraffin wax in the fuel begins to come out of solution (Cloud Point); any further cooling will cause wax to separate out of solution. The temperature at which a certain fuel will become saturated with wax and causes filter plugging problems is termed the Cold Filter Plugging Point (ASTM D6371). The temperature at which fuel will no longer flow is the Pour Point (ASTM D97). At the pour point, most of the fuel is still liquid, although it is very thick or viscous and trapped in a honeycomb-like network of wax crystals.

Since diesel powered equipment is frequently used at temperatures low enough to cause wax to separate, a number of techniques have been devised to prevent the wax from causing problems by plugging fuel screens, lines, filter, and so on, and preventing fuel flow to the engine. Vehicles

designed to operate at very low temperatures have provisions for heated fuel tanks, insulated fuel lines, heated fuel filters and other mechanisms to warm the fuel so that the wax does **not** separate. These more elaborate systems are usually **not** practical in more temperate climates where they are needed **only** a few days a year.

Fuel Filters

Fuel filters have already been discussed in detail in the Fuel Filters section of this bulletin. The **only** additional consideration in terms of common issues with winter fuels is that using a large filter or multiple filters in parallel will allow more fuel wax to be filtered before a power loss occurs. Also, relocating the fuel lines and filter out of the wind-stream and wheel splash and into the engine compartment near the engine block will help keep them warm.

Engine Idling

CAUTION

Do not idle the engine for excessively long periods of time with engine coolant temperature below the minimum specification found in the applicable engine Owner's Manual. This can result in fuel dilution of the lubricating oil, carbon build up in the cylinder, cylinder head valve sticking, and/or reduced performance.

Additives

There are a number of fuel additives available which reduce the pour point and cold filter plugging point (CFPP) of diesel fuel. These are commonly referred to as pour point depressant additives, cold flow improver additives, wax crystal modifiers, or fluidity improver additives (and can be collectively termed "Winter Additives"). Certain additives can reduce the Pour Point by as much as 21°C [70°F] and the CFPP by as much as -1°C [30°F]. A survey of winter blend fuels by the Bureau of Mines (now a part of the Energy Research and Development Administration) revealed that a large percentage of the commercially marketed diesel fuels had been treated with a winter additive. Before purchasing such an additive to treat fuel, ask the fuel supplier whether the fuel already contains a winter additive. Depending on the amount and type of additive already in the fuel, additional additives will or will **not** be necessary.

These additives alter the size and shape of wax crystals, allowing pumping of fuel at lower temperatures. Although certain additives can be very effective, they are **not** a cure all. Their performance varies depending on the paraffin type and content of the fuel treated. Severe weather applications can require fuel warmers in addition to additives. Although other additives are available that can provide some benefits, PowerService Diesel Fuel Supplement® + Cetane Boost® is the **only** fuel additive endorsed by Cummins Inc. to help prevent filter gelling in cold weather applications.

Fuel Warmers

Warming diesel fuel just prior to filtration is an excellent method of preventing fuel filter plugging. If cold fuel is warmed sufficiently, the wax crystals will dissolve in the fuel. The dissolving requires warming to a temperature of approximately 11 to 22°C or 20 to 40°F above the fuel's cold filter plugging point.

In order for a fuel warmer to reliably prevent fuel filter plugging due to wax, it **must** be capable of supplying enough heat to the fuel at the maximum fuel flow (**not** just fuel consumption) rate to raise the fuel temperature from the lowest expected fuel temperature (probably the lowest expected ambient temperature) to 11 to 22°C or 20 to 40°F above the fuel's cold filter plugging point. There are four different fuel warmers presently offered by Cummins Filtration™ to raise the temperature of the inlet fuel.

- Fuel Filter Heater - The Cummins Filtration™ Positive Temperature Coefficient (PTC) fuel filter heats the fuel before the fuel flows into the fuel filter. The heater is installed on the fuel filter head. Most complaints of fuel waxing occur in the fuel filter. The heater uses ceramic discs that sense the fuel temperature and heat the fuel to a temperature just above the cloud point.

The PTC heater is self-regulating. Depending on battery voltage, the heaters use from 6 to 25 amperes at maximum output. When no heat is required, the heater uses less than 0.5 ampere. The heater can be left on during engine operation or it can be turned off with the cab switch. The heater reaches full heating capacity in about two minutes. The PTC heater kit is available (see Table 13, Fuel Filter Heaters).

Table 13, Fuel Filter Heaters

Watts	Cummins Filtration™ Part Number
300	3836029-S

A Cummins Filtration™ Kit, Part Number 3837317-S, adapts the heater to most fuel filter heads with 1 in-14 threads. Use the following fuel filter list to identify fuel filter heads with 1 in-14 threads. The heater adds about 25 mm [1 in] in height to the fuel filter head assembly.

Fuel Filter List with 1 in -14 Threads	
FF-104	FF-213
FF-105	FF-105C
FS-1242(B)	FS-1001
FS-1000	FS-1212
FF-105D	FS-1003

The Cummins Filtration™ Kit, Part Number 3832054-S, adapts to FS-1251 filter.

2. Recirculating Fuel Warmer, Part Number 3305782, can be used to warm inlet fuel for flow up to 9.5 l/pm [2.5 gpm]. The unit circulates engine coolant around the inlet fuel to warm the fuel. The unit is most effective when immersion or tank heaters are used to warm the coolant. An optional thermostat, Part Number 3305783, can be used to bypass fuel when 27°C [81°F] is reached. Use Table 14, Temperature Rise Chart to determine the performance capability of this fuel warmer for different operating conditions.

Recirculating Fuel Warmer, Part Number 3305782 - Performance Data

Table 14, Temperature Rise Chart.

Inlet Fuel Temperature	Outlet Fuel Temperature Fuel Flow Rate		
	2-1/2 GPM	1-1/2 GPM	1/2 GPM
-34°C [-30°F]	0°C [32°F]	3°C [38°F]	11°C [52°F]
-23°C [-10°F]	4°C [39°F]	7°C [45°F]	13°C [55°F]
-12°C [10°F]	8°C [47°F]	12°C [53°F]	15°C [59°F]
-1°C [30°F]	16°C [60°F]	17°C [62°F]	19°C [67°F]
10°C [50°F]	22°C [71°F]	23°C [74°F]	25°C [77°F]
21°C [70°F]	29°C [85°F]	31°C [87°F]	31°C [88°F]

3. Thermo Blend™ - The Cummins Filtration™ Thermo Blend™ fuel warmer recirculates warm deaerated drain fuel from the engine to the filter and injection system, rather than allowing it to return to the tank. A 10 to 15 minute engine-running warm up period is usually necessary to provide successful operation. A built-in thermostat automatically bypasses fuel at 43°C [109°F]. Part Number 3310200 **must** be used for all Midrange and Heavy Duty diesel engines. Part Number 3308750 **must** be used for all heavy duty off-highway equipment (such as 12 and 16 cylinder engines).

4. Thermo Blend™ FM, Part Number 3310630, The Cummins Filtration™ Thermo Blend™ FM fuel warmer, combines the return fuel heating principle with a special filter head. When used with Cummins® Part Number 3315843 (Cummins Filtration™ Part Number FS-1212) fuel-water separator, it provides fuel dewaxing, water removal, and filtration. A built-in thermostat automatically bypasses fuel at 21°C [70°F]. When using fuel warmers, do **not** overheat the fuel. The maximum fuel temperature at the inlet to the fuel pump is 70°C [158°F]. Alterations of heating devices **must** be reversible, or have some means to turn them off during warm weather operation. The fuel tank is heated by the injector return (drain) fuel from the engine. On typical installations, the cooling effect of the tank maintains fuel temperatures at an acceptable level.

On some installations, such as acoustically enclosed units, little cooling of the tank occurs because of the design. On these installations, a fuel oil cooler can be used to limit the temperature of the fuel at the fuel pump inlet to 70°C [158°F] or less.

Depending on the particular engine model involved, the engine horsepower will begin to decrease slightly above fuel inlet temperatures of 46°C [115°F]. The percent of power loss is **not** as great on engines with the Cummins® PT™ and HPI® fuel system (less than 1 percent per 5°C or 9°F), due to the inherent viscosity compensating characteristics (see Power Loss section in this bulletin). Operation above 70°C [158°F] is **not** recommended due to the loss of the lubricating quality of the fuel with resultant wear to the fuel system components which depend on fuel for lubrication. A fuel warmer will **not** help if the fuel is below the pour point and can **not** be pumped to the warmer; therefore, in extremely cold conditions, fuel can be treated with light distillate fuel or treated with a pour point depressant to reduce the pour point, or it can be necessary to heat the fuel to allow it to flow.

When using fuel warmers that use engine coolant as a source of heat, some form of coolant heating during shutoff will allow the heater to become effective much more quickly after start-up. These fuel warmers **must** also be checked for leaks. Since the fuel warmer is on the suction side

of the fuel pump and the cooling system is pressurized, any small leak will allow coolant to enter the fuel system.

Other Considerations

Wax in the fuel will deposit in any restriction or sharp bend in the fuel plumbing system. If fuel starvation occurs during cold-weather operation and plugged fuel filters are **not** found, look for plugging of tank pick-up screens, sharp bends in the fuel lines, fittings, and so forth.

Water Contamination

Free water (non-dissolved) in the fuel can freeze at low temperatures and the resulting ice crystals can plug fuel filters causing fuel starvation. Care **must** be taken to keep fuel storage tanks dry. Tanks can be “stuck” often with water detecting paste (usually obtainable from fuel suppliers) to be sure they are dry. If water is detected, it **must** be pumped out.

Keeping bulk fuel storage tanks dry has already been mentioned; however, if this is a persistent issue, a dryer (fuel-water separator) can be installed on the bulk fuel dispensing system.

Condensation in the vehicle fuel tank(s) occurs when the air in the fuel tank(s) cools down during a shutdown period. This moisture can be reduced by filling the vehicle fuel tank before engine shutdown to reduce the air space above the fuel.

Dissolved water comes out of solution as fuel cools. As fuel cools from 4 to -29°C, [39 to -20°F] the solubility of water in the fuel reduces 70 percent. Therefore, fuel pumped from a relatively warm underground tank into a vehicle which sits overnight in sub-zero temperatures can cause some free water to separate. However, this source of free water is almost negligible, because even at high temperatures, fuel will dissolve very little water (0.1 mass-percent at 71°C) [160°F].

Note : More cold weather engine operation recommendations are in Service Bulletin Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html) and in the engine operation and maintenance manual.

Microbial Contamination of Diesel Fuel

⚠ WARNING ⚠

Although most of the microbes that will live in fuel tanks are common organisms to which humans are constantly exposed, contact with microbes or fungi from a fuel tank must be avoided. When a fuel system is contaminated and cleaning is necessary, workers must be protected. Remember that the fungi produce reproductive spores and when dry these can easily become airborne, so breathing protection must be provided or the microorganisms must be kept wet. Dispose of the water and sludge removed from fuel tanks properly. Never place these materials in sanitary sewer system since they can kill bacteria used in sewage treatment. Never place them in storm sewers or surface water streams since they can kill fish and other aquatic animals.

⚠ WARNING ⚠

The most common problem associated with exposure to these microbes is dermatitis which in some people can be quite serious. Any exposed skin must be thoroughly washed with warm, soapy water.

⚠ WARNING ⚠

Avoid eating, drinking and smoking while working with these microbes. Any ingestion of the microbes or exposure to broken skin must be considered serious. It is recommended that if this happens the worker be taken to a doctor, along with a sample of the microbes.

⚠ WARNING ⚠

Biocides are generally only mildly toxic to humans and animals but must still be handled carefully. In cases of ingestion or contact with the eye, follow manufacturer's recommendations. Seek medical attention.

This section covers the recognition of and solutions to microbial contamination of diesel fuel.

To protect against fuel shortages, many users have been storing fuel and, as a result, the frequency of microbial contamination has increased. Microbial contamination of fuel, though **not** a new concern, is more common in metalworking industries which use water-soluble oils as cutting fluids or in long-term storage of hydrocarbon fuels than it is in diesel fleet operations. All hydrocarbon fuels are essentially sterilized by the high temperatures encountered in the refining process; however, they can become contaminated soon after leaving the refinery by microorganisms. These microorganisms, primarily bacteria and fungi, exist rather harmlessly in moisture-free fuel, passing through fuel systems without having any negative effects.

However, in the presence of water, these microorganisms begin to grow and reproduce. The rate of growth depends on how well the environment suits the particular microorganism's needs.

The growth of a large colony of microorganisms in a fuel system can cause several issues. The first and usually most obvious is fuel filter plugging with a greenish-black or brown slime, frequently accompanied by a foul odor. This slimy, string-like colony can also plug sharp bends in fuel lines, fuel meters, and other restrictions. The second issue these microorganisms can cause is corrosion due to the acid by-products some of them produce. It is also possible, if the microorganisms pass through the fuel filter, that they will form deposits and cause damage in the fuel pump and injectors.

Some indicators of microbial contamination are:

- Slime deposits on tank walls, piping, or other surfaces which are exposed to fuel. These deposits are usually greenish-black or brown and are slick to the touch.
- Black or brown "stringy" material suspended in tank water bottoms.
- Swelling or blistering of any rubber surface (washers, hoses, connectors, and so forth) that comes in contact with fuel.
- Sludge or slime deposits on filter surfaces.
- Foul odor resembling that of rotten eggs (hydrogen sulfide).

The following list shows test kits of which we are aware. Listing of a kit can **not** be construed as a recommendation or approval; and, the fact that a kit is **not** listed **only** means we are unaware of it. Cummins Inc. has **not** tested any of these kits, but has **only** reviewed the manufacturer's literature. Users **must** evaluate the kits available to them and select one based on their own judgment.

- Total Count Sampler, Catalog Number MTOO 000 25 for package of 25, from Millipore Corporation, Bedford, MA 01730, 1-800-645-5476. The Total Count Sampler contains a nutrient media specifically designed to encourage bacterial growth; however, many fungi will grow on it. Millipore recommends incubation at 35°C [95°F] for 24 hours; however, they can be incubated at room temperature for 36 to 48 hours. If the results on the Total Count Sampler are low and are still suspected, re-sample using Millipore™ Yeast and Mold Sampler (Catalog Number MYOO 000 25 for package of 25). This sampler contains a nutrient media which suppresses the growth of most bacteria, but is rich in nutrients for fungi. For best results, use both samplers each time water bottoms are tested. These Millipore™ samplers are probably the most sensitive of those listed, and in fact, can lead to overtreatment of a fuel system. Millipore™ samplers are also available from Millipore in Australia, Belgium, Brazil, Canada, Denmark, England, Finland, France, Italy, Japan, Mexico, Norway, Spain, Sweden, Switzerland, and West Germany. Inquiries from other countries can be directed to Millipore Intertech, Inc., P.O. Box 255, Bedford, MA 01730 U.S.A.
- Microbe Monitor Test Kit (From Air BP®) British Petroleum Cleveland-Hopkins International Airport Cleveland, OH 44135, 1-800-533-2340. One sample per kit. Incubates at room temperature.

When it has been established that microbial contamination is present and action **must** be taken, there are several approaches. The most obvious solution is prevention. Most of the bacteria and fungi involved are soil organisms which can become airborne or waterborne. Prevention of the entrance of microorganisms is **not** possible because these organisms can enter the fuel through many different routes.

Growth of these microorganisms can be prevented. Since all metabolic processes of an organism are conducted in water, denying the microorganism access to water will prevent growth, thus preventing the development of large, troublesome colonies. Therefore, the first and most important step in prevention is to keep fuel systems dry. Keeping a fuel system entirely dry is impossible. In cases where microbial contamination is a recurring issue, a microbicide can be used to chemically treat the fuel or the water.

There are three general classes of biocides: water-soluble, fuel-soluble, and universally soluble. Fuel-soluble biocides are best suited for treating fuels which are to pass through several storage steps in the distribution process. A fuel-soluble biocide injected into the fuel early in the distribution system is carried with the fuel through the entire downstream system, effectively sterilizing the fuel until usage. Fuel-soluble biocides are easier to add to the fuel system since the exact amount needed to treat a volume of fuel is easily determined and they have a low toxicity to human and other life forms. The obvious disadvantage to fuel-soluble biocides is cost; each batch of new fuel added to the system **must** be treated since the biocide is consumed as the fuel is consumed.

Water-soluble biocides are more economical for use in treating one step in a fuel distribution system, such as the end-user's storage tank. The water-soluble biocides, since they are insoluble in fuel, stay where they are placed until the water bottoms are pumped from the tank; therefore, the total amount of biocide purchased is less. There are a number of disadvantages to water soluble biocides. Since no biocide is carried downstream by the fuel, each successive tank in the system **must** be individually treated. There is some difficulty in determining how much biocide to place in a tank, since that depends on how much water is in the tank. The biocide can **not** be thoroughly mixed with the water in the bottom of a tank. Water-soluble biocides are much more easily taken in by humans and other life forms: and therefore, **must** be disposed of properly when water bottoms are pumped from a tank. Water bottoms containing a water-soluble biocide **must not** be placed in a sanitary sewer system because the biocide can destroy the bacteria used by sewage treatment plants. These water bottoms **must** be treated as an acidic, industrial oily waste.

Universally soluble biocides are soluble in both water and diesel fuel. They allow you to treat the entire downstream system. However, each subsequent load of fuel does **not** need to be treated. The biocide will remain in any water that has collected at the bottom of the storage tank and continue to inhibit microbial growth. With certain types of biocides, the interval between treatments can be as long as six months. Like water-soluble biocides, universally soluble biocides are more easily taken up by humans and other forms of life. They also tend to be more expensive than the other types of biocides.

Treating a fuel tank that is infested with a large population of microorganisms will kill the microorganisms, but it will **not** eliminate the filter plugging they can be causing. The water and sludge containing the microorganisms **must** be removed from the fuel systems. First, clean the fuel system thoroughly. Next, a fuel-soluble or universally-soluble biocide **must** be added to the next few batches of fuel to kill any remaining microorganisms. Finally, the addition of a water-soluble or universally-soluble biocide can be continued for at least several months to be sure the microorganisms are all dead. If microbial contamination is a recurring issue, it is recommended that use of the water-soluble or universally-soluble biocide be continued permanently, since this will be the most cost effective solution to the problem. This can be done by determining the amount of water that accumulates in tank bottoms between pump outs and adding about double or triple the amount of water-soluble biocide recommended to treat that volume of water. For example: you normally pump out approximately 379 liters [100 gallons] of water bottoms. In this case, after pumping the bottoms, immediately add to the fuel tank two or three times the amount of biocide normally used to treat 379 liters [100 gallons] of water. Since the biocide is more dense than fuel, it will settle to the bottom of the tank and will dissolve in the water as it accumulates. After refilling a fuel tank, the biocide **must** be allowed to settle before drawing fuel from it to keep from pumping the biocide out with the fuel.

Fuel Discoloration (Black Fuel)

In some Cummins® engines, normal operation can cause the diesel fuel in the engine and fuel tank to appear dark or black in color. Discoloration of the fuel can be caused by the following: engine lubricating oil mixing with the fuel during operation, the formation of asphaltenes, degradation of fuel from storage, or from the practice of blending lubricating oil with fuel to be burned by the engine.

Mixing of engine lubricating oil

In some fuel systems, engine lubricating oil and diesel fuel are used in close proximity to each other for lubricating and sealing purposes. This interface is a function of fuel pump and/or injector design. Under certain operating conditions, a small amount of lubricating oil can mix with the diesel fuel and be returned to the tank, causing the fuel to appear dark in color. It takes a very small amount of lubricating oil (less than 0.1 percent) to cause the fuel to become visibly darkened. This small amount of oil in the fuel will have no adverse affects on engine performance, durability, or reliability.

Asphaltene formation

This phenomenon is common for fuel systems that operate at very high fuel pressures and temperatures. The high temperature fuel that is **not** injected into a combustion cylinder is returned to the fuel tank. As the fuel is recirculated and exposed to the same high pressures and temperatures during continuous operation, asphaltenes begin to form larger clusters of insoluble materials that can lead to fuel discoloration. If these formations of asphaltenes grow large enough, they will be captured by the filter element and can lead to high filter restriction and shortened filter life.

Asphaltene formation due to fuel overheating can be aggravated by a lack of fuel coolers, improperly sized fuel tanks, inadequate mixing of return fuel between tanks, low fuel levels, or installation of the fuel tank that prevents dissipation of heat. If an engine or fuel system cooling issue is suspected, the cooling systems **must** be inspected and modified as necessary to comply with Cummins Inc. requirements. Existing fuel filters could possibly need to be resized or additional filtration could possibly need to be added to increase contaminant holding capacity. Consult a Cummins® Authorized Repair Location for guidance in fuel filter selection and cooling system issues.

Degradation of fuel from storage

Refer to the “Microbial Contamination of Diesel Fuel” section of this service bulletin.

Darkening of diesel fuel due to the mixing of lubricating oil or asphaltene formation does **not** indicate a manufacturing defect or other warrantable malfunction, and is a function of normal operation. Customers should continue using the equipment as is, unless the discoloration has resulted from fuel overheating, fuel contamination, or if fuel filter life is being reduced and causing operational difficulties with the equipment.